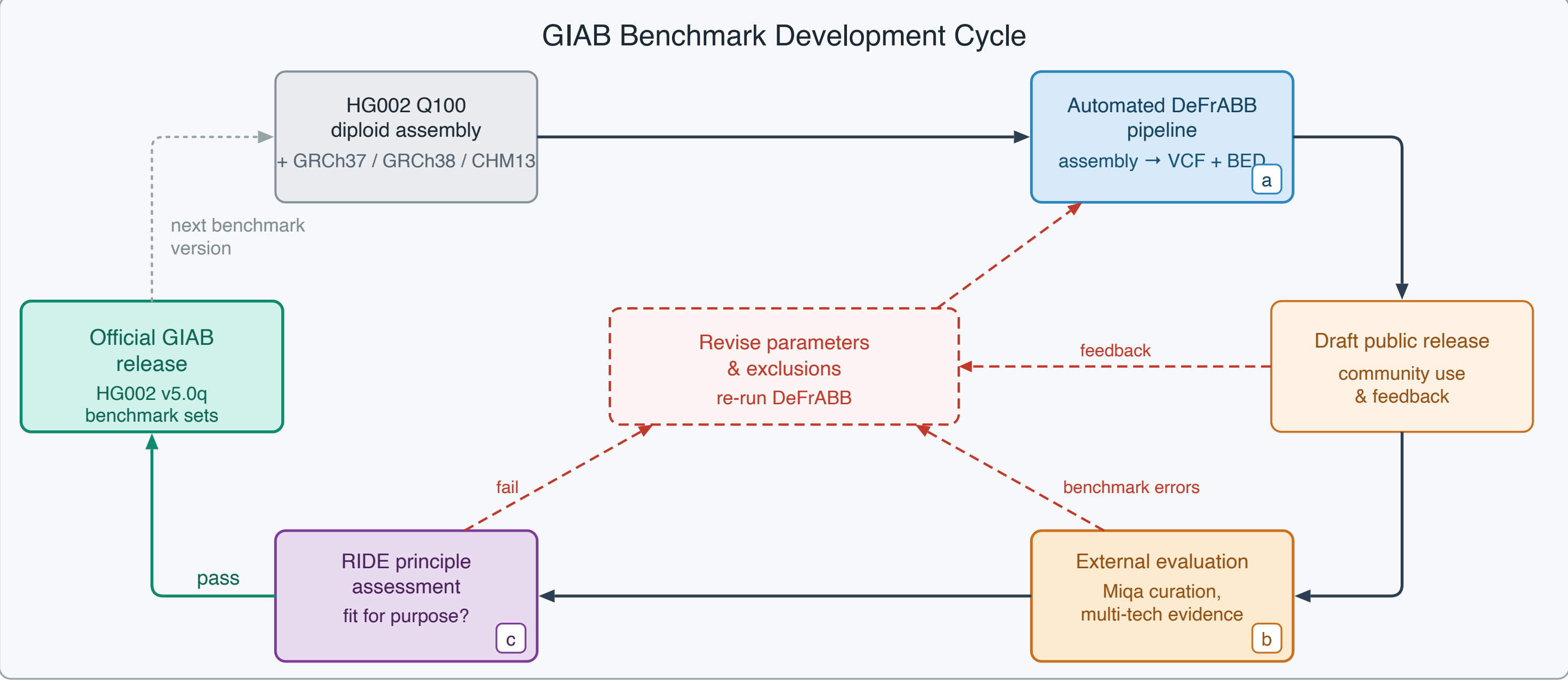


GIAB Benchmark Development Cycle



a DeFrABB pipeline

1 Assembly-based variant calling

Mat.

Ref.

Pat.

■ Large SV segment: present in only one sequence
▵ Ribbon narrows where region sizes differ (INS/DEL)

Ribbons connect matched regions; large INS/DELS change region size between assembly and reference
Output: variant calls (VCF) + diploid regions (BED)

2 Draft benchmark generation

Phased variants (SNVs, indels, SVs — all sizes)

Diploid regions (1:1 alignment, both haplotypes)

Exclusions (gaps, repeat breaks, complex SVs, errors)

Benchmark = diploid regions – exclusions

Separate small-var + SV sets · VCF + BED each x3 references (GRCh37, GRCh38, CHM13v2.0)

3 Evaluation — internal QC

Compare draft to known callsets → analysis report
→ optimize parameters & re-run (rapid iteration)

Draft benchmark sets
small-variant + SV · VCF + BED · x3 references
→ external evaluation, panel (b)

b External evaluation & curation

Draft benchmark sets — from panel (a)
VCF + BED · small-var + SV · x3 references

External comparison callsets
Evaluators call variants from own sequencing data:

Illumina	short read
PacBio HiFi	long read
Oxford Nanopore (ONT)	long read
Element Biosciences	short read
Ultima	short read

Small-variant + SV callsets from external groups

Compare benchmark vs callsets
hap.py / vcfeval (small variants) · truvari (SVs)
comparisons on GRCh38

Discrepancies (potential FP / FN)

Miqa curation (stratified random subset)

- Pre-configured IGV sessions: GRCh38 + Q100 haplotypes
- Multi-tech read evidence: Illumina, Element, HiFi, Onso, ONT-UL
- Curated by submitting group
- Classify: correct/incorrect/unsure

✓ ✗

Incorrect / unsure
NIST re-curation;
errors → cycle hub

Benchmark correct
discrepancy = error in
comparison callset

c RIDE principle

Is the benchmark fit for purpose?

Reliable Identification of Errors (RIDE)
Olson, Nat. Rev. Genet. 24, 2023

A benchmark is fit for purpose when discrepancies with high-quality comparison callsets are reliably errors in the callsets — not the benchmark.

Benchmark vs comparison callset

Within benchmark regions (BED)

Discrepancies (FNs + FPs) → curated in panel (b)

Curated outcomes decide fitness

■ benchmark correct — error in comparison callset
■ benchmark incorrect
■ unsure

✓ **Fit for purpose — pass**
Most discrepancies are errors in the comparison callsets, not the benchmark

✗ **Not fit — fail**
Benchmark errors common; revise parameters & re-run DeFrABB — red dashed path in cycle